

BEAUTY TECH PRESENTATION

LUMÉA

Glow, your way

A Modern Single-Product Beauty E-Commerce Platform

React 18 · Node.js Express · MySQL · Tailwind CSS

Project by: Kichu Khoirom (University Roll No. 2433548)

Degree: Bachelor of Computer Applications (BCA)

Institution: Gulzar Group of Institutions, Khanna, Punjab

Industry Internship: Viro Web Solution (Web Development)



The Product — Luméa Glow Tint



Multifunctional Formula

A silky cream tint engineered to melt seamlessly into the skin for an effortless, buildable flush on cheeks, lips, and eyelids. Replaces 3 cosmetic products in 1 pocket-sized 8g component.



Clean & Conscious Beauty

Formulated with 100% vegan botanical waxes, cruelty-free certification, zero mineral oils, and enriched with hydrating jojoba and squalane for 12-hour continuous skin dewiness.



Approachable Luxury (₹799)

Direct-to-consumer pricing with zero retail middleman markups. Ships free across India with Cash on Delivery (COD) support and a guaranteed 4.9★ customer satisfaction rating.

FORMULA & TEXTURE

Formulated for skin that looks like skin

Pure botanical waxes infused with hydrating spheres for weightless color that melts into cheeks and lips.

Product Specifications

✓ Product Type	Multifunctional Cream Blush & Lip Tint
✓ Net Weight	8g / 0.28 oz
✓ Finish	Deerly Natural Glow
✓ Coverage	Sheer to Medium Buildable
✓ Suitable For	All Skin Types (Sensitive, Dry, Combination, Oily)
✓ Key Ingredients	Botanical Plant Waxes, Vitamin E, Hyaluronic Spheres
✓ Formulation	100% Vegan, Cruelty-Free, Paraben-Free, Fragrance-Free
✓ Shelf Life	24 Months
✓ Origin	Formulated in India

Curated Color Palette — 6 Signature Shades



Rose Petal

#D9828B · Soft Rosy Pink

Universal everyday blush for cool to neutral undertones



Peach Bloom

#EFA07F · Warm Coral Peach

Sun-kissed radiance ideal for light to medium warm skin



Berry Kiss

#A94B68 · Deep Rich Plum

Bold romantic statement tint for medium to deep undertones



Soft Coral

#E87970 · Fresh Coral Tangerine

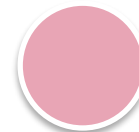
Lively playful flush perfect for daytime lip and cheek glow



Nude Glow

#B97862 · Warm Terracotta Nude

Subtle sculpted neutral giving natural 'no-makeup' definition



Pink Champagne

#E8A5B5 · Shimmer Mauve Pink

Dewy highlighted glow with ultra-fine light-reflecting mica

Market Challenge vs. The Luméa Experience

✗ Traditional Cosmetic E-Commerce

- Catalog Clutter: Overwhelming marketplaces with hundreds of unrelated products cause severe decision fatigue.
- Shade Hesitation: Static studio photos don't demonstrate shade warmth or skin undertone matching.
- Painful Checkout: Mandatory multi-page logins, hidden shipping fees, and broken UPI options lead to 70%+ cart drop-offs.
- Post-Purchase Anxiety: No live tracking visibility once an order is confirmed.

✦ The Luméa Single-Product Experience

- Laser-Focused D2C Storytelling: One hero formula perfected across 6 universally flattering shades.
- Live Swatch Synchronizer: Clicking any shade instantly switches photography, ambient glow, and inventory count.
- Frictionless Checkout: 1-click bag drawer, transparent pricing (₹799 with Free Shipping), and reliable Cash on Delivery.
- Total Transparency: Real-time 5-stage fulfillment stepper + self-service order cancellation & ticket desk.

Full-Stack Technology Architecture

⚡ React 18 + Vite 5.4

Single-Page Application frontend with declarative component hierarchy, custom context hooks (Cart, Auth, Notifications), and sub-second HMR updates.

⚡ Tailwind CSS 3.4

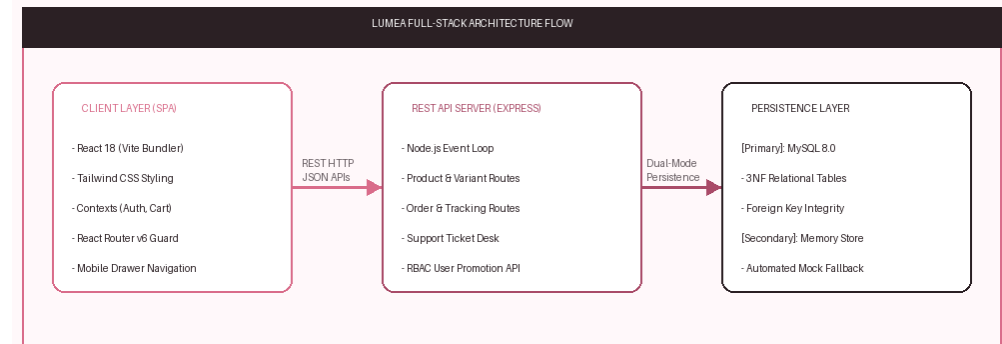
Utility-first design system with customized tokens (#FFF8FA, #D96C8A, #2B2024), 60 FPS mobile transforms, and zero layout drag.

⚡ Node.js & Express REST API

Asynchronous non-blocking backend server orchestrating products, orders, support tickets, and role verification.

⚡ MySQL 8.0 + In-Memory Fallback

Relational database in 3NF with automated zero-downtime fallback to an in-memory replica if database is unconfigured.



Interactive Storefront & Swatch Experience

Customer Journey Highlights:

1. Real-Time Swatch Switching:

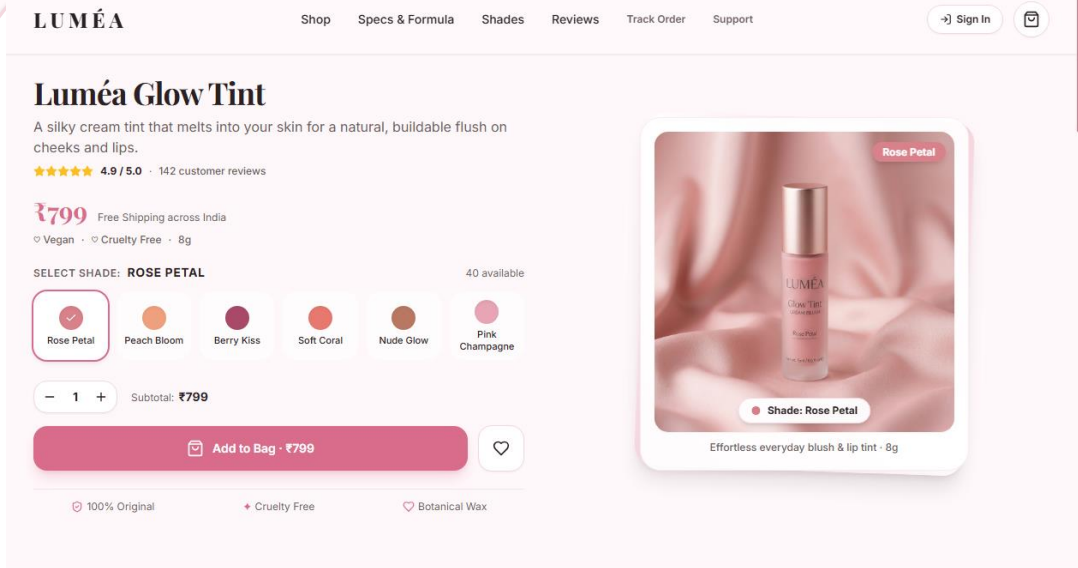
Selecting any color swatch instantly updates the product photography, switches the ambient background glow, and renders stock availability.

2. Formula & Benefits Breakdown:

Dedicated sections showcase clean botanical ingredients (jojoba, squalane), cruelty-free pledge, and customer testimonials.

3. Persistent Cart Drawer:

An off-canvas bag slides into view with live quantity controls, subtotal calculations, and free delivery indicators across India.



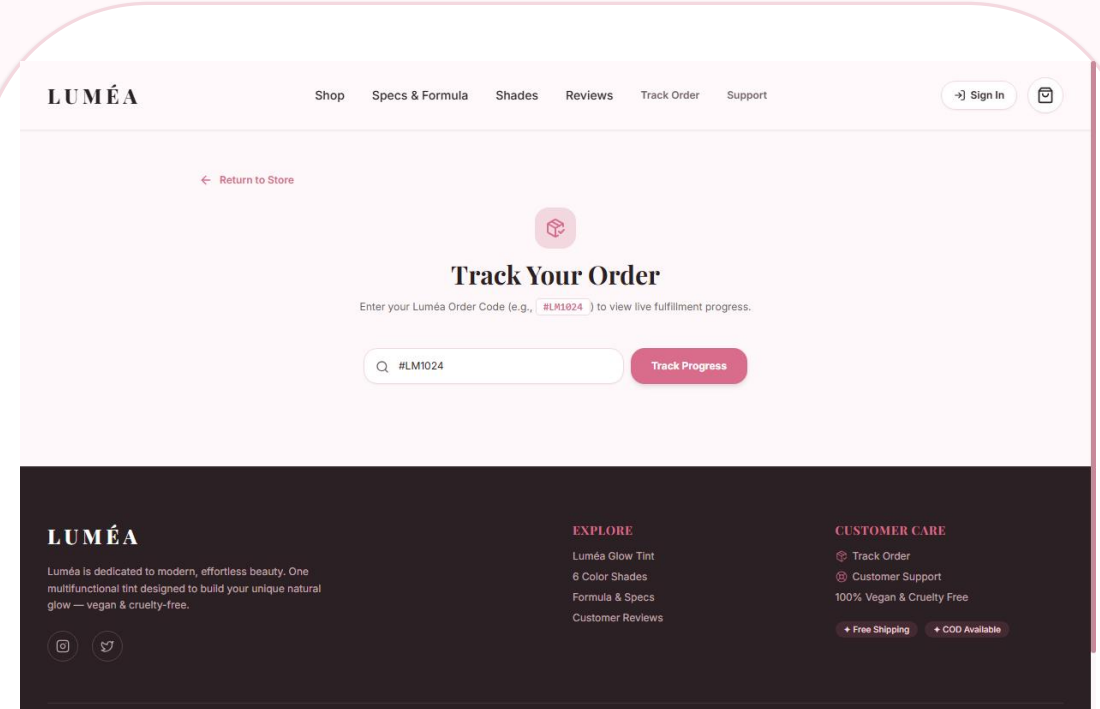
Checkout, Fulfillment & Order Tracking

Frictionless Checkout Flow

- Validates name, phone, city, and shipping address.
- Zero surprise fees: transparent ₹799 subtotal with free shipping.
- Active Cash on Delivery (COD) + friendly maintenance banner for UPI online payments.

5-Stage Visual Order Stepper

- Search by order code (e.g. #LM1024) to inspect live progress.
- Visual Stepper: Pending → Confirmed → Processing → Shipped → Delivered.
- Customer Self-Cancellation: Allowed during pending state; restores inventory units automatically.



Administrative Control & RBAC Governance

Role-Based Access Hierarchy:

1. Super Admin (kichuKhoirom@gmail.com):

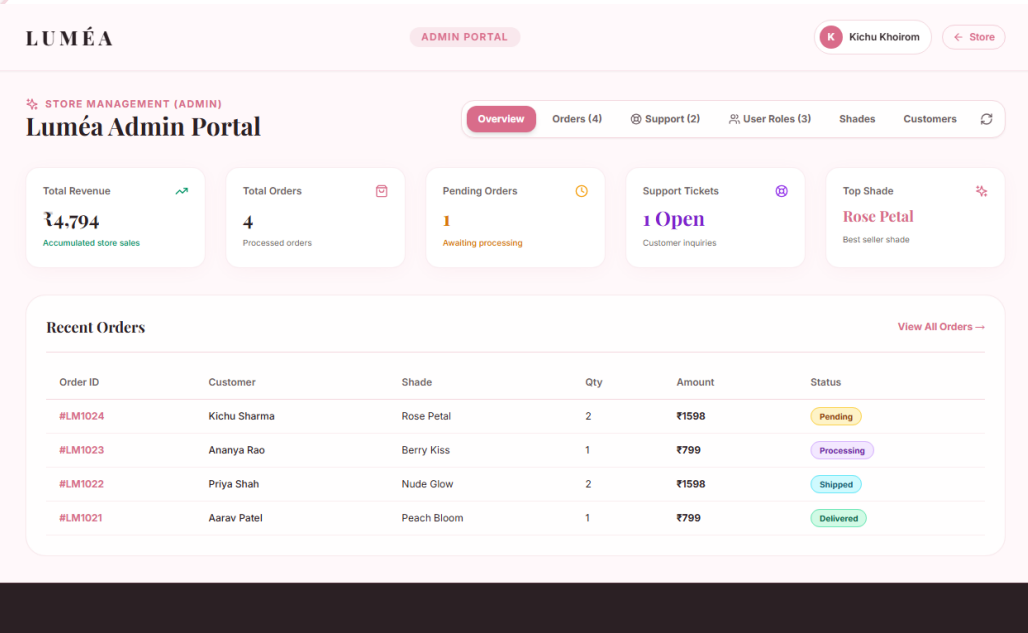
- Full administrative rights.
- Promote customers to Sub-Admin, demote roles, delete users.
- Access real-time revenue KPIs and support tickets.

2. Sub-Admin (pardeepsign@gmail.com):

- Order status management (update to Confirmed/Shipped).
- Customer support ticket resolution.
- Product inventory inspection.

3. Customer (james@gmail.com):

- Storefront shopping, order tracking, and ticket submission.
- Access to /admin is strictly blocked by Route Guard.



Performance Profiling & Fault Tolerance

⚡ 60 FPS Mobile Optimization

Heavyweight CSS backdrop-filter blurs were eliminated from general cards. On screens $\leq 640\text{px}$, blurs are disabled entirely; on desktop, blur is isolated to the sticky header and hardware-accelerated with `translateZ(0)`. Zero jank or drag.

🛡️ Multi-Tier Error Containment

Custom 404 Page renders a luxury gradient numeral with store recovery CTAs; dedicated `/503` page offers retry connectivity; root-level React `ErrorBoundary` prevents blank-screen whiteouts.

🔄 Zero-Downtime Database Fallback

If the local MySQL database is offline or unconfigured, the backend automatically transitions to an in-memory replica pre-loaded with products, variants, sample orders, and user roles.



Loading Luméa Beauty...

Future Roadmap & Project Conclusion

Live Payment Gateway Integration

Direct integration with Razorpay and Cashfree webhooks for instant UPI QR code generation, credit/debit cards, and automated payment reconciliation.

AI-Powered Webcam Shade Matcher

Client-side computer vision engine using TensorFlow.js face landmark detection to analyze skin undertones in natural lighting and recommend the ideal tint shade.

Automated WhatsApp & SMS Alerts

Instant order dispatch notifications, live delivery agent tracking links, and review requests powered by Gupshup / Twilio webhook messaging.

Regional Language Localization

Multilingual store localization (Hindi, Punjabi, English) to dramatically expand reach across diverse beauty customer segments in India.